

# Forensic Report - Fingerprint Comparison

Fingerprint Analysis Workstation - Version 2.3

Report generated at: 2026-05-22 14:14:03

**Lab disclaimer:** This report contains statistical software outputs and must not be used as sole identity evidence.

**Audit Trail**

**Case Reference:** -

**Operator:** -

**SHA-256 (Reference file):**  
d61fa650ac6b127531ff88c5646682945450e3056cf255cec24737f356aba604

**SHA-256 (Query file):**  
c02bdd597340fb459d97eafe38375c73c512854d55e277b6e10357ebb16286ac

## Methodology Summary

Grayscale conversion, size normalization, CLAHE, optional denoising, Otsu/adaptive binarization, multi-orientation Gabor ridge enhancement, thinning, minutiae extraction (ending/bifurcation) with filtering, and one-to-one constrained matching by distance/angle/type. For partial-print verification, the query minutiae are searched with translation and small rotation inside the reference frame.

### Recorded Run Parameters

| Parameter     | Value |
|---------------|-------|
| border_margin | 12    |
| min_distance  | 14    |
| min_contrast  | 20    |

|                              |             |
|------------------------------|-------------|
| min_angle_diff               | 12          |
| denoise_method               | fastNlMeans |
| fast_denoise_h               | 10          |
| gauss_ksize                  | 3           |
| original_zoom                | 100         |
| partial_zoom                 | 50          |
| partial_shift_x              | 0           |
| partial_shift_y              | 0           |
| apply_preview_scale          | True        |
| auto_scale_normalization     | True        |
| auto_scale_factor_applied    | 1.4         |
| MATCH_DISTANCE_THRESHOLD     | 15          |
| MATCH_ANGLE_THRESHOLD_DEG    | 35          |
| PARTIAL_VERIFY_SEARCH_RADIUS | 80          |
| PARTIAL_VERIFY_STEP_PX       | 10          |

Pipeline Images

Reference Processing Pipeline

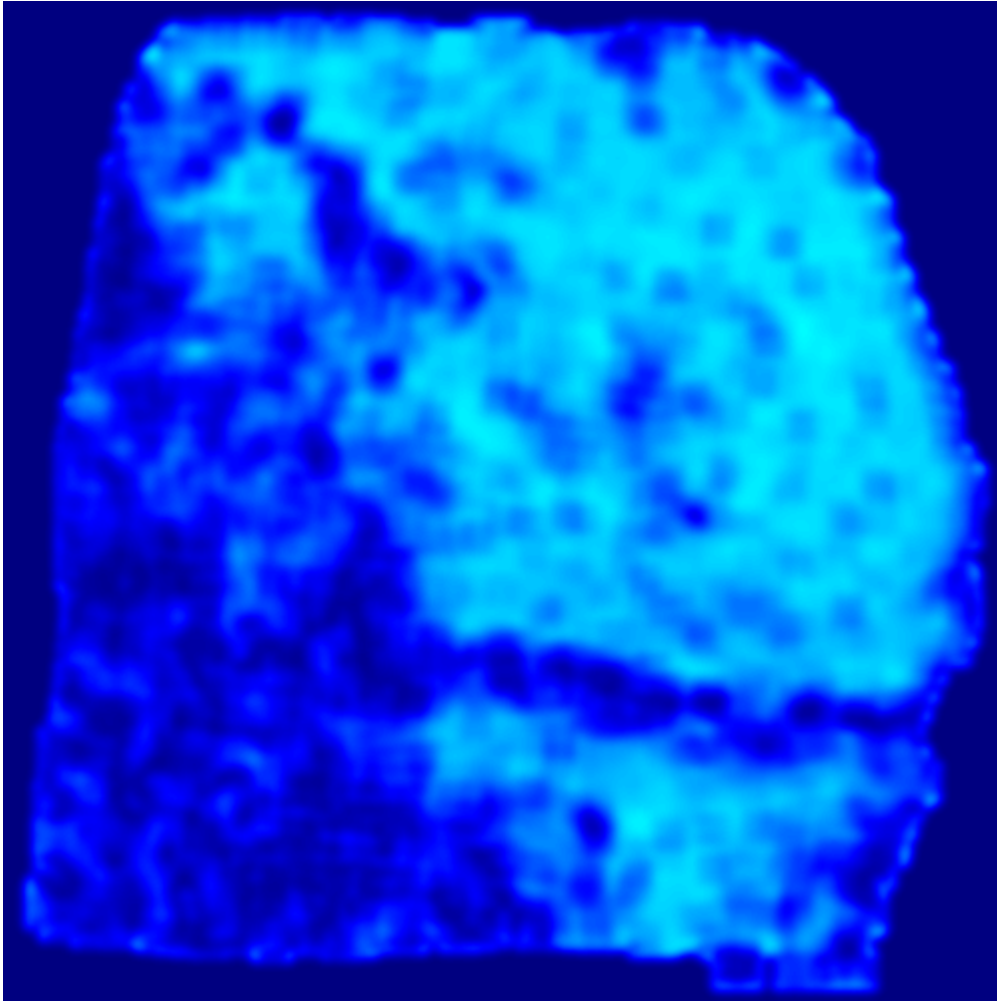
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**Stats:** Binary white pixels: 85316 | Gabor white pixels: 30473 | Skeleton white pixels: 16202 | Filtered minutiae: 87

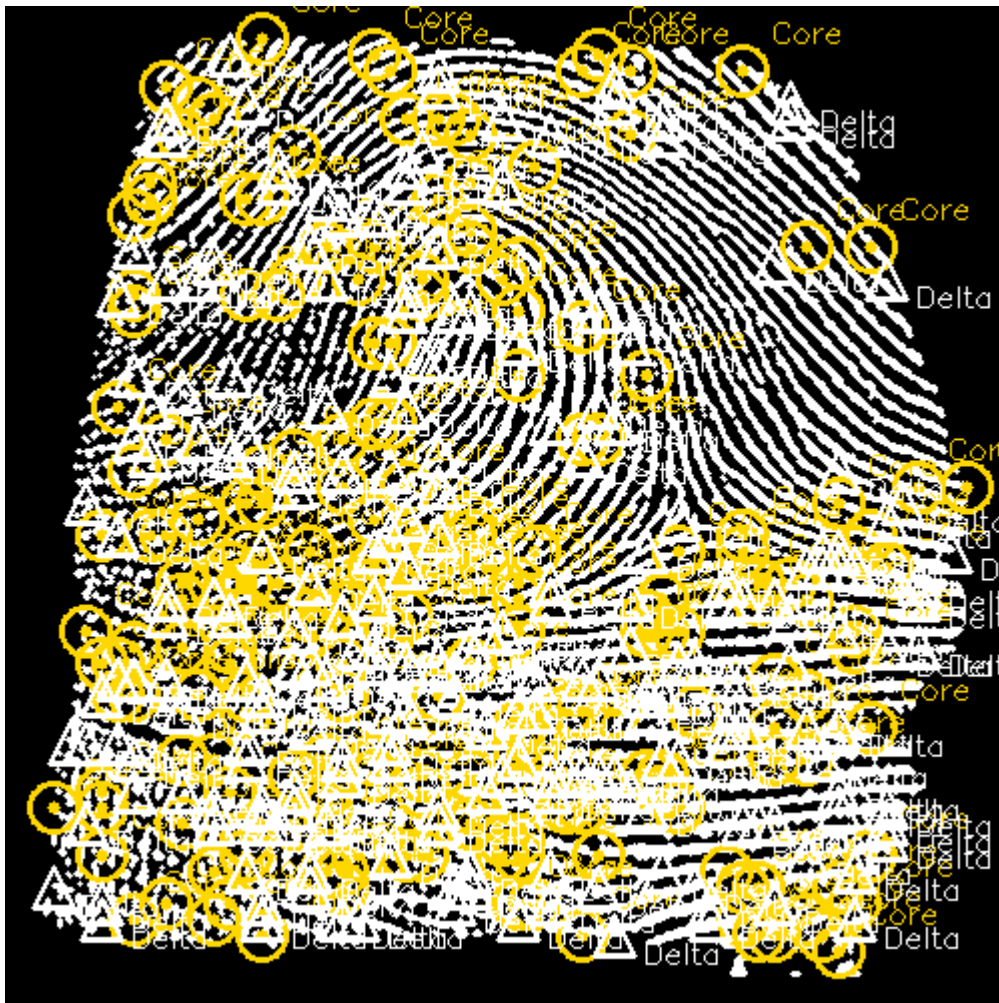
Preprocessed Binary



Quality Heatmap



Singular Points



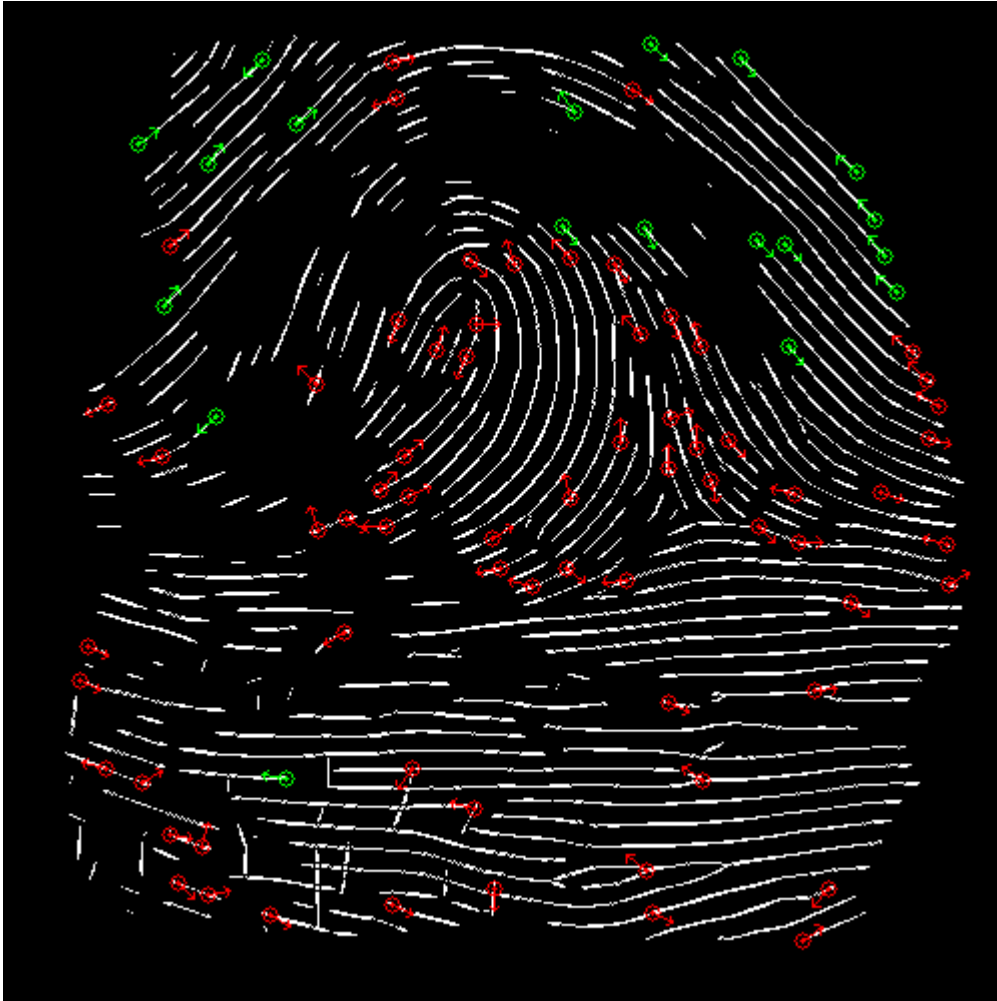
Ridge Enhancement (Gabor)



Skeleton



Minutiae Visualization



## Query Processing Pipeline

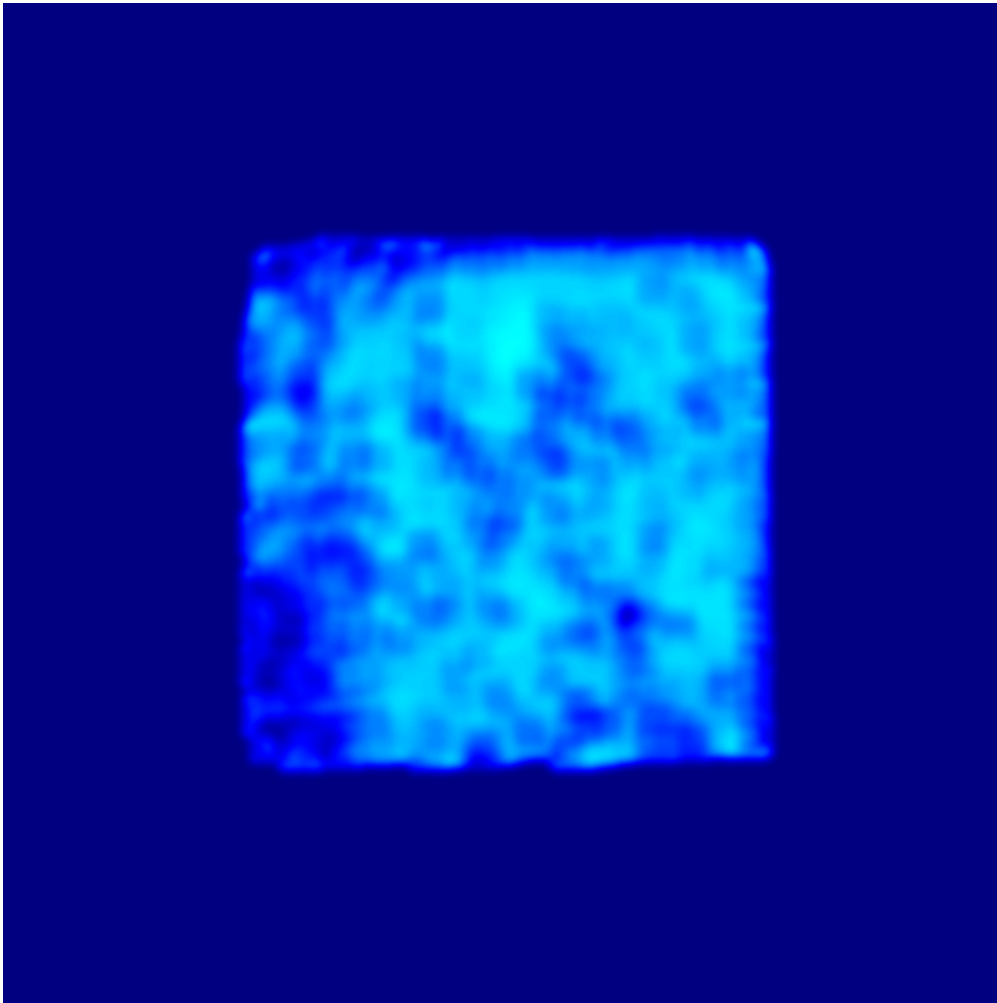
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**Stats:** Binary white pixels: 29297 | Gabor white pixels: 14262 | Skeleton white pixels: 6407 | Filtered minutiae: 47

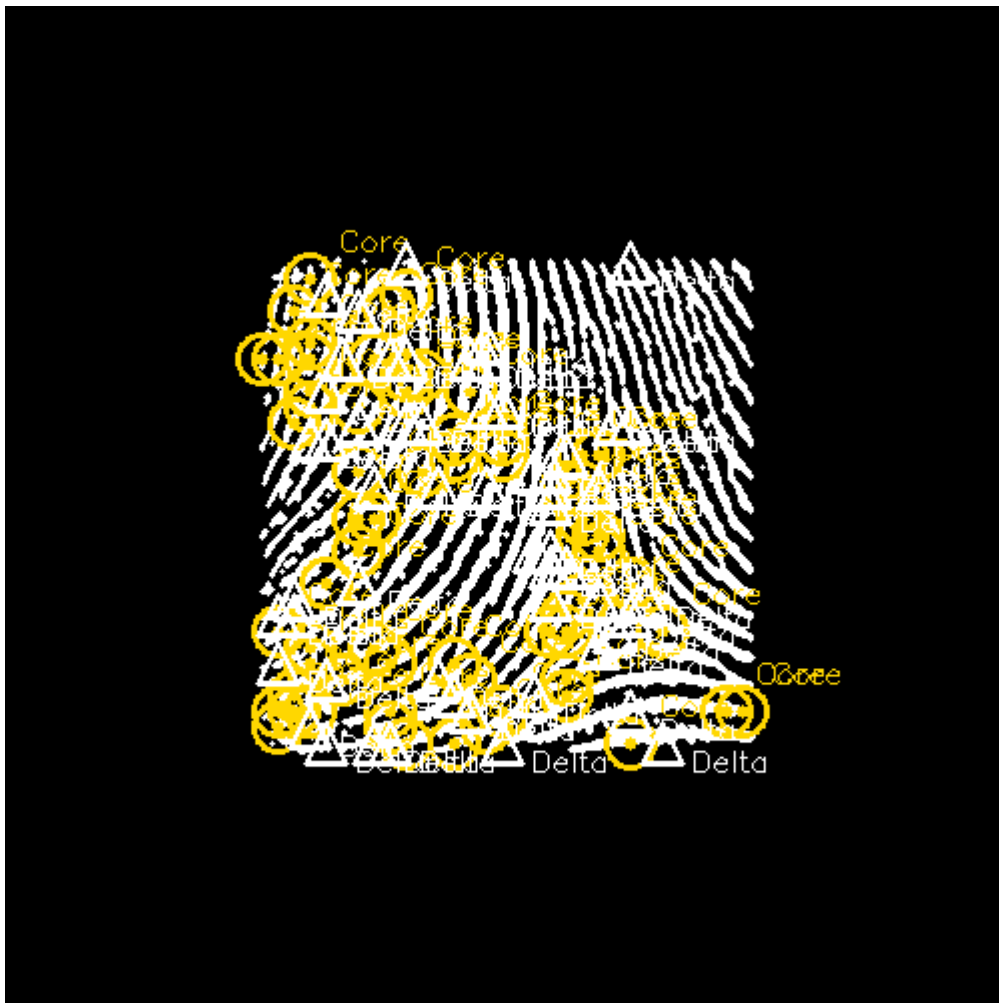
Preprocessed Binary



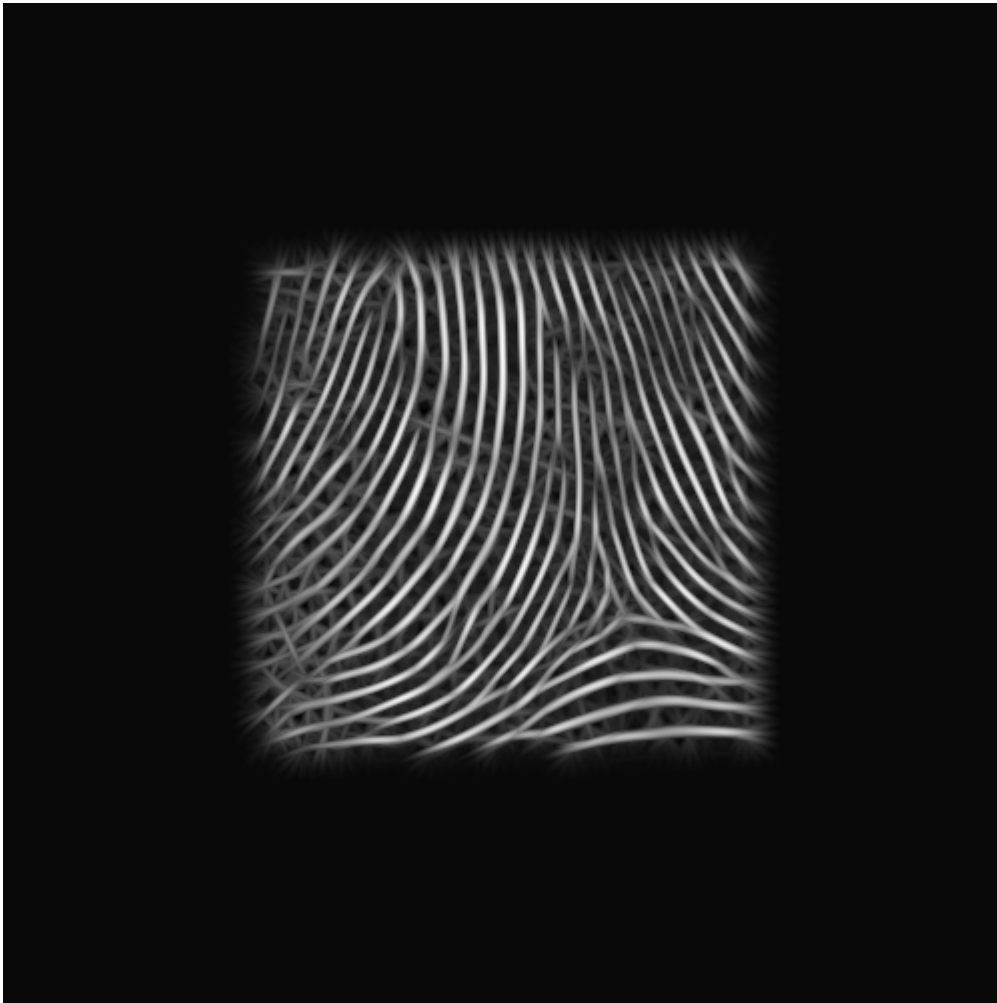
Quality Heatmap



Singular Points



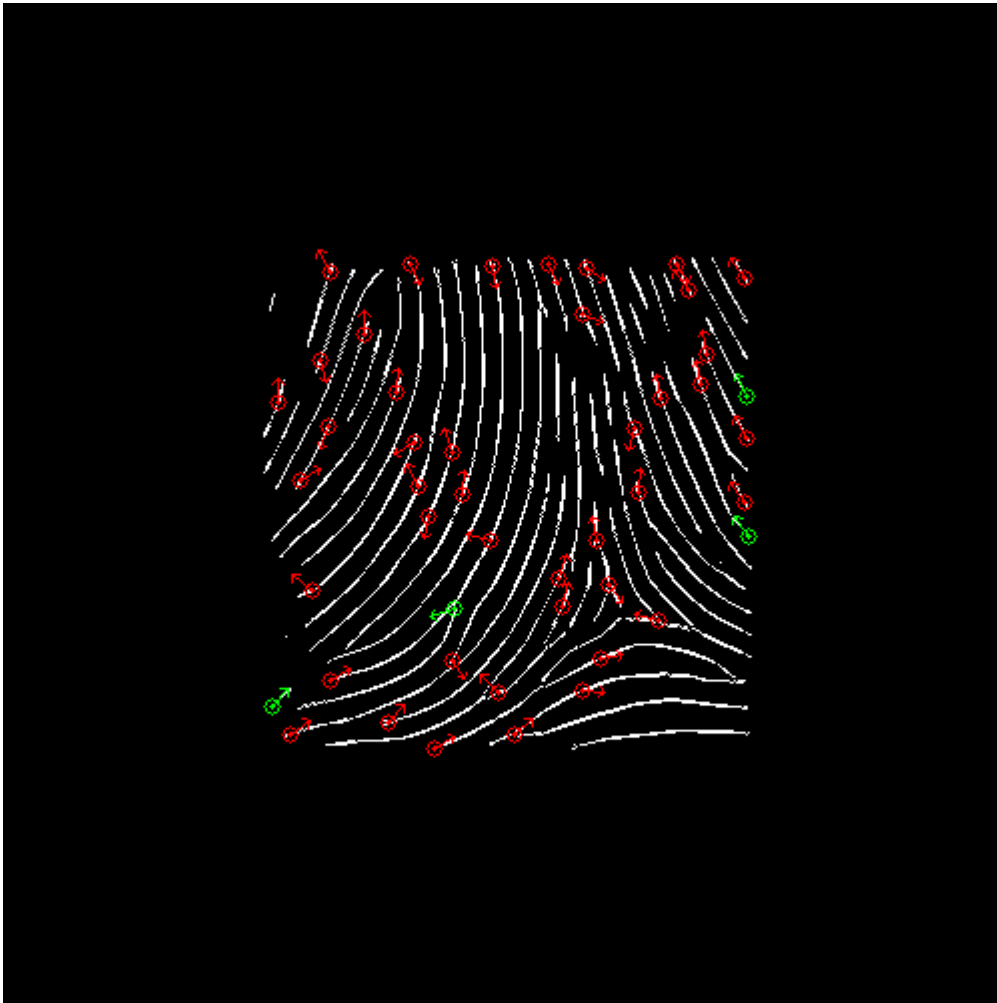
Ridge Enhancement (Gabor)



Skeleton



Minutiae Visualization



## Aligned Match Visualization

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Match Overlay



## ORB Visual Verification

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### ORB Feature Matches



Final Skeletons Used for Matching

Reference Skeleton



Query Skeleton



Archive PNG files: original\_skeleton.png, partial\_skeleton.png in report folder.

**Partial-to-Reference Alignment**

Approximate alignment: dx=42 dy=-41 rot=-19.0 deg. Baseline matches: 3, final matches: 11, gain: 8, score delta: +17.02.

Green: matched reference points | Orange: matched partial points after alignment.

**Statistical Comparison Results**

Reference minutiae count: 87

Query minutiae count: 47

One-to-one matched pairs: 11

Match score (query-based): 23.40%

Fused Score: 32.58%

Combined verdict: Weak or inconclusive similarity — expert review advised

Decision mode: partial-first

Fusion Components - Minutiae: 23.40% | MCC: 44.23% | ORB: 0.60%

Score explanation: 11 / 47 = matched pairs divided by total extracted query minutiae. This is not area coverage.

Dice score: 16.42%

Forensic tier: Tier C - Low statistical similarity under configured thresholds

This system reports statistical metrics for fingerprint representation comparison under recorded parameters. It must not be used as sole identity proof and requires qualified expert review and lab SOP.

Internal status: LOW MATCH